M. Sc. Physics (Semester-II) 2026
Quantum Mechanics II – Tutorial 5

1. Evaluate the matrix elements (or expectation values) given below. If any vanishes, explain why it vanishes using Wigner Eckart theorem

   (a)$\langle n=2, \ell=1, m=0|x|n=2, \ell=0, m=0\rangle$

   (b)$\langle n=2, \ell=1, m=0|p_z|n=2, \ell,=0, m=0\rangle$

   Here, $|n\ell, m\rangle$ stands for the energy eigenket of a nonrelativistic hydrogen atom with spin ignored.

2. Consider the matrix elements $\langle n'l'm'_\ell m'_s|(3z^2-r^2)|n\ell m_\ell m_s\rangle$ and $\langle n'\ell'm'_\ell m'_s|xy|n'\ell'm'_\ell m'_s\rangle$ of a one-electron atom.

   (a) Identify the operators whose common eigenstate is $|n\ell m_\ell m_s\rangle$.

   (b) What physical quantity do operators $3z^2-r^2$ and xy represents?

   (c)Write the selection rules of $\Delta\ell$, Δm_ℓ, and Δm_s. Justify your answer. (Hint: Express $3z^2-r^2$ and xy as component of tensor operator using $Y_\ell^m(\theta,\phi)$ and then use the Wigner Eckart theorem.

   Useful formulae: $Y_2^{\pm 2} = \sqrt{\frac{15}{32\pi}}\frac{(x\pm iy)^2}{r^2}$, $Y_2^0 = \sqrt{\frac{5}{16\pi}}\frac{3z^2-r^2}{r^2}$.

3. Recall that finite rotation is obtained through $exp(\frac{i}{\hbar}J.n\theta)$. Consider finite rotation by angle θ around z axis in spin -1/2 representation.

   (a) Obtain the rotation matrix through exponentiation of appropriate matrix. You should obtain: $\cos\frac{\theta}{2}\mathbb{I} + i\sin\frac{\theta}{2}\sigma_z$

   (b) What happens to a state in spin-1/2 representation when you apply a 2π rotation?

   (c) Calculate how the spin operator $\vec{S}=(S_x, S_y, S_z)$ transforms under rotation. (Hint: if state transforms by $U|\Psi\rangle$, operators transform as $UOU^\dagger$. Substitute expression in (a) for U and use expression you obtained in your H. W. for S_x, S_y, S_z in terms of Pauli matrices)